Selection of Motor Power Rating

The power rating of a motor for a specific application must be carefully chosen to achieve economy with reliability.

Use of a motor having insufficient rating may lead to:

1- fail to drive the load.

2- lower the productivity and reliability through frequent damages and shut-downs due to overloading of the motor and power modulator.

3- use of motor having extra power than needed.

4 - extra initial cost.

5 - extra loss of energy.

6- operate below rated power factor especially, IMs & Synch. Motors**.**

- It is clear that, heat is produced due to losses in copper, iron and friction during motor operation.
- As the temperature of the motor increases beyond ambient value part of the heat flows to the surrounding medium.
- As outflow of heat increases, equilibrium sets when the heat generated becomes equal to heat dissipated.
- The motor temperature reaches steady state and it depends on power loss, which intern depends on the output power of the machine; and this dependence of temperature rise on output power which is taken as thermal loading on the machine.
- Steady state temperature is not the same at various parts of the machine. It is usually highest in the winding because loss density in conductors is high and dissipation rate is low. (since windings are embedded in slots)**.**
- Among the various materials used in machines, insulation has the lowest temperature limit.

The temperature rise which electric motors may safely withstand is determined by the thermal stability (limiting temperature) of the insulating materials used in them. By using materials of higher thermal stability, motors of the same physical size may be rated for greater power output. Fuller utilization of the motor is also attained through better cooling. According to their limiting temperature all insulating materials employed in electric machines are divided into the following seven classes:

*1.**Class Y insulation:*** cotton yarn(خيوط) and fabrics(اقمشة), paper and fibrous materials of cellulose or silk neither impregnated (مشربة)with nor immersed in a liquid dielectric. The limiting temperature for this class of materials is 90°C.

*2. **Class A insulation:*** The same materials as above, but impregnated with, or immersed in a liquid dielectric. For this class the limiting temperature is 105°C.

*3. **Class E insulation*:** Certain synthetic organic films (اغشية عضوية اصطناعية معينة) and other materials having the same thermal stability. The limiting temperature is 120°C.

*4. **Class B insulation***: mica-, asbestos-, or glass-fiber-base materials with an organic binding agent; the limiting temperature is 130°C. (الميكا- أو الأسبستوس- أو المواد ذات القاعدة من الألياف الزجاجية مع عامل ربط عضوي)

*5. **Class F insulation:*** mica-, asbestos-, or glass-fiber-base materials combined with suitable synthetic binding and impregnating agents; the limiting temperature is 155°C.

*6. **Class H insulation:*** mica, asbestos or glass fiber combined with silicone binding and impregnating agents; the limiting temperature is 180°C.

*7. **Class C insulation:*** mica, ceramic materials, glass, or quartz used without or with inorganic binding agents. For this class of materials, the limiting temperature exceeds 180°C and is restricted only by the specific physical, chemical or electrical properties of the material concerned.

The temperature limits for various classes can be summarized in the following table:

Insulation class	Y	A	E	B	F	H	C

Temperature limit, °C	90	105	120	130	155	180	above 180

The limiting temperatures should never be exceeded in service because it will result in deterioration and breakdown of the insulation and will shorten the service life of the motors.

As concerns checking of the motor for its overload capacity and starting torque, this should be done regardless of the results of the calculation, based on motor heating. *The following table* gives the coefficients used to determine the permissible instantaneous overload torque of different electric motors.

Table ( ): Motor Instantaneous Torque Overload Coefficients ($\lambda=T_{max}/T_n$)

Type	***Torque overload coefficient***
DC Motors	2 (for special types up to 3 - 4)
Slip-Ring Motors	2-2.5
General Purpose Squirrel-Cage Motors	1.8 - 3
Double and Deep-Bar Squirrel-Cage Motors	1.8 - 2. 7
Synchronous Motors	2 - 2 .5 (for special types up to 3 - 4)
AC Commutator Motors	2 - 3

Determination of Motor rating

• To determine motor rating, duty cycles discussed previously can be broadly classified as: -

i) Continuous duty
ii) Fluctuating loads
iii) Short-time and intermittent duty

i) Continuous duty: -

• The selection of the motor capacity for such a drive is simple if we know approximately the constant power input required by the machine.

• In order to select a motor for a rating equal to the known power input requirement, one may be sure that this rating is the maximum permissible with respect to heating because the manufacturer designs and rates it so as to attain maximum utilization of the material used at rated output.

• For continuous duty at constant or slightly varying load, a motor is selected from the relevant catalogue for a capacity corresponding to the power required.

• Although losses during starting are greater than those under rated load, they may be neglected because starting under continuous duty is very infrequent (نادر)and practically no influence on motor heating.

1- When taking into account the static (steady-state) mode and neglecting the dynamic state, i.e. considering the dynamic moment as zero.

$T= T_L$, $T_{dyn} =0$

The motor is selected with the specified values (Nominal or Rating parameters) according to the load power referred to the motor shaft and at the angular speed value in the equivalent system, as follows:

$\omega_n=\omega$, $P_n=kP_{eq}$, where, $P_{eq}=\omega_n$, k=(1-1.3)-safety factor.

1- When taking into account the static and dynamic states. The motion equation is:

$T =T_L+ T_{dyn}$

You must know the speed profile, which is the relationship of speed change with time, and it consists of the following parts: 1-Starting interval, 2 - Constant speed running interval , 3- Braking Interval and 4- Idling interval. We must know the load profile or loading diagram, which is the relationship of the change in the load torque with time. An example illustrates how to calculate equivalent motor torque and equivalent motor power.

Data: -

1- $\omega=f(t)$; Speed profile with numerical values.

2- T_L= f(t) ; Loading diagram with numerical values.

3- J_{Σ}=0.5kg.m^2 ; Total moment of total inertia.

Required:

1- Plot electric motor torque profile and electric motor output power profile.

2- Calculate electric motor equivalent output power.

Solution:

1. Motor torque calculation

$T = T_L + T_{dyn} =$

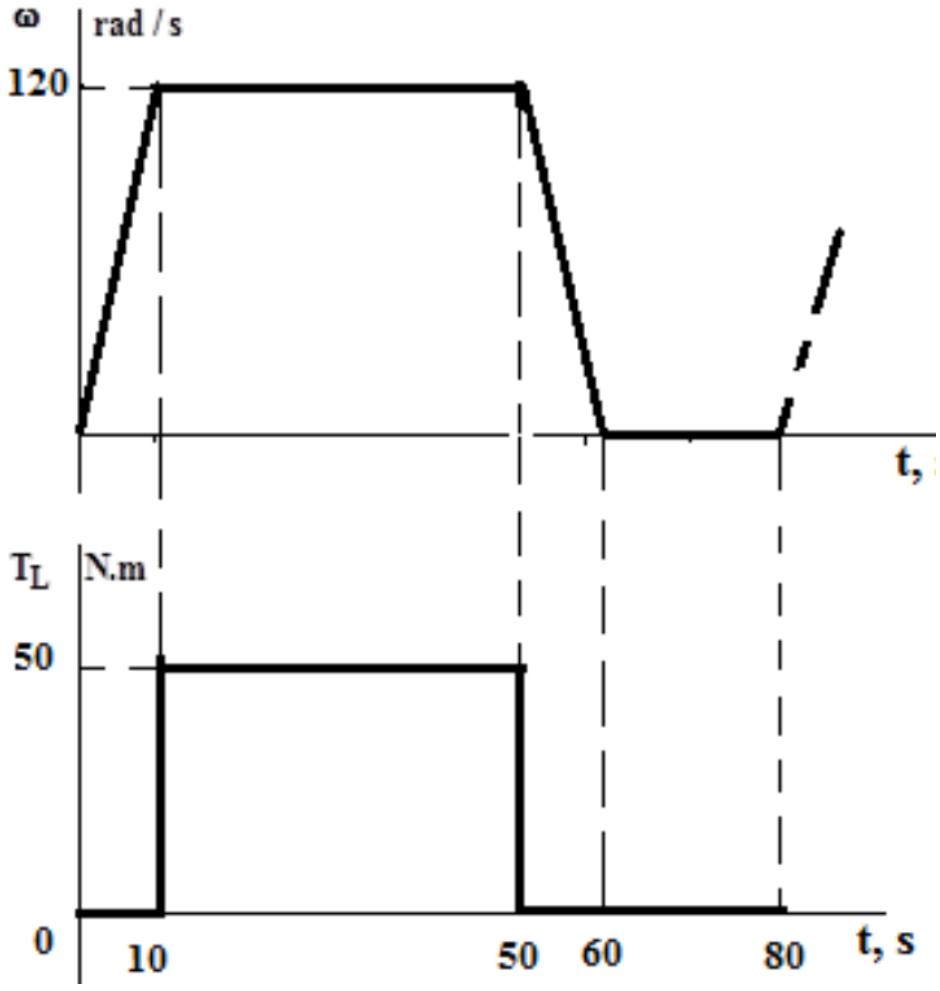
$$T_{dyn} = J_{\Sigma}\frac{d\omega}{dt} = J_{\Sigma}\frac{\Delta\omega}{\Delta t}$$

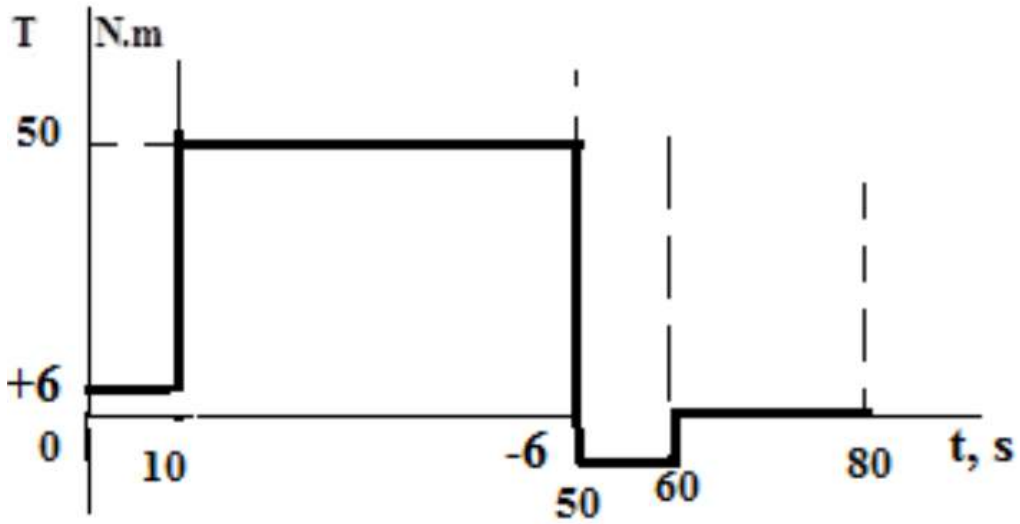
t=0- 10 s ,

$$T_{dyn} = 0.5\frac{120}{10} = +6 \text{ N.m}, T = 6 \text{N.m}$$

t= 10- 50 s, $T_{dyn} = 0, T = +50 \text{N.m}$

t= 50-60 s, $$T_{dyn} = 0.5\frac{-120}{10} = -6 \text{ N.m}, T = -6 \text{N.m}$$





2. Motor output power calculation

$$P = T\omega \text{ , watt}$$

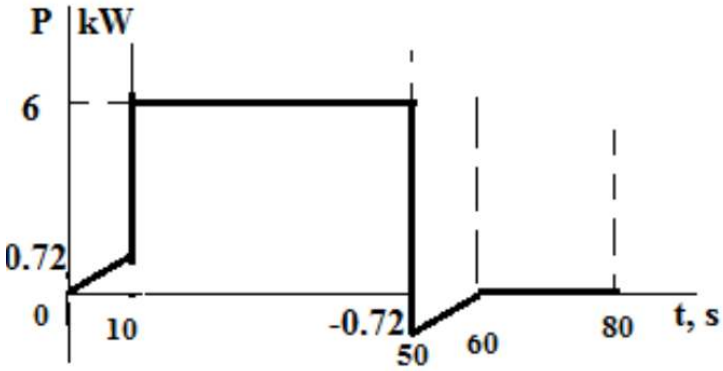
$$t = 0 - 10 \text{ s}, P = +6\frac{120}{10}t = +0.072t \text{ kW}$$

$$t = 10 - 50 \text{ s}, P = 50 * 120 = +6 \text{ kW}$$

$$t = 50 - 60 \text{ s}, P = T\omega_o + T\frac{\Delta\omega}{\Delta t}t = -6(120)+(-6(-\frac{120}{10}t) = -0.72 + 0.072t \text{ kW}$$

$$t = 60 - 80 \text{ s}, P = 0$$

Equivalent power calculation:



$$P_{eq} = \sqrt{\frac{\int_0^{10}(0.072t)^2 dt + \int_{10}^{50} 6^2 dt + \int_{50}^{60}(-0.72+0.072t)^2 dt + 0}{80}} = 4.247\,kW$$

ii) Fluctuating and intermittent duty

- Equivalent current, torque and power methods - This method is based on approximation that the actual variable motor current can be replaced by an equivalent Ieq , which produces same losses in the motor as actual current. The equivalent current is determined as follows;
- As we know, motor loss (ΔP_l) consists of two components.
- P_c - Constant loss which is independent of load (Core loss and friction loss)
- P_{cu} - Variable loss dependent on load (copper loss)

Thus for a fluctuating load consisting of n values of motor currents (I_1 , I_2 , I_3 , ….., In) for duration t_1 , t_2 , t_3 , …., t_n respectively, the equivalent current method states that : variable motor

current may be replaced by an equivalent current I_{eq} produces the same losses in the motor as the actual current. So: -

$$P_c + I^2_{eq}R = \frac{\left(P_c + I_1^2 \ R\right)t_1 + \left(P_c + I_2^2 \ R\right)t_2 + \ldots + \left(P_c + I^2{}_n R\right)t_n}{t_1 + t_2 + \ldots + t_n}$$

$$P_c + I^2_{eq}R = \frac{P_c\left(t_1 + t_2 + \ldots + t_n\right)}{t_1 + t_2 + \ldots + t_n} + \frac{\left(I^2{}_1 t_1 + I^2{}_2 t_2 + \ldots + I^2{}_n t_n\right)R}{t_1 + t_2 + \ldots + t_n}$$

$$I_{eq} = \sqrt{\frac{I^2{}_1 t_1 + I^2{}_2 t_2 + \ldots + I^2{}_n t_n}{t_1 + t_2 + \ldots + t_n}}$$

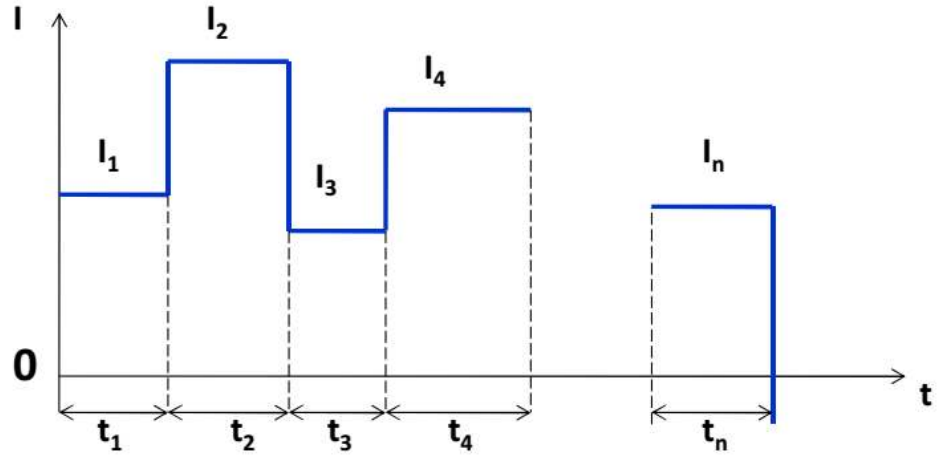


Fig. Load diagram of fluctuation load

- If the current varies smoothly over a period T, Ieq is: -

$$I_{eq} = \sqrt{\frac{1}{T}\int_0^T i^2 dt}$$

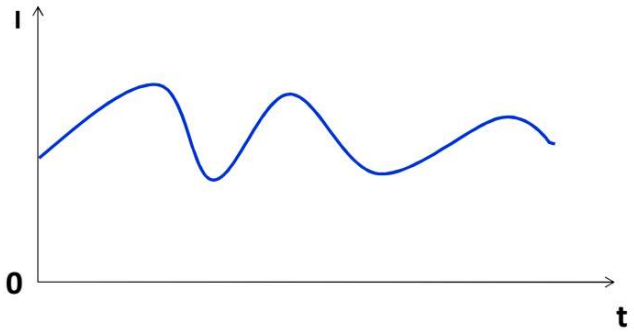


Fig. () Load diagram of fluctuating load

- After I_{eq} is determined, a motor with next higher current rating from commercially available rating is selected $I_n \geq I_{eq}$. Then this rating is checked for its practical feasibility (الجدوى العملية)

$I_{Lmax}/I_n \leq \lambda_I$. For general purpose DC motors λ_I=2-2.5 where for in induction motors λ_I=1.65-2.75. $\lambda_I = I_{max}/I_n$.

If the load torque diagram is given, then the motor can be selected according to the equivalent torque T_{eq}

$$T_{eq} = \sqrt{\frac{T_1^2 t_1 + T_2^2 t_2 + \cdots + T_n^2 t_n}{t_1 + t_2 + \cdots + t_n}}$$

The motor is selected with $T_n \geq T_{eq}$ then checked for $T_{Lmax} / T_n \leq \lambda_T$
If the load power diagram is given, then the motor can be selected according to the equivalent power P_{eq}

$$P_{eq} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + \cdots + P_n^2 t_n}{t_1 + t_2 + \cdots + t_n}}$$

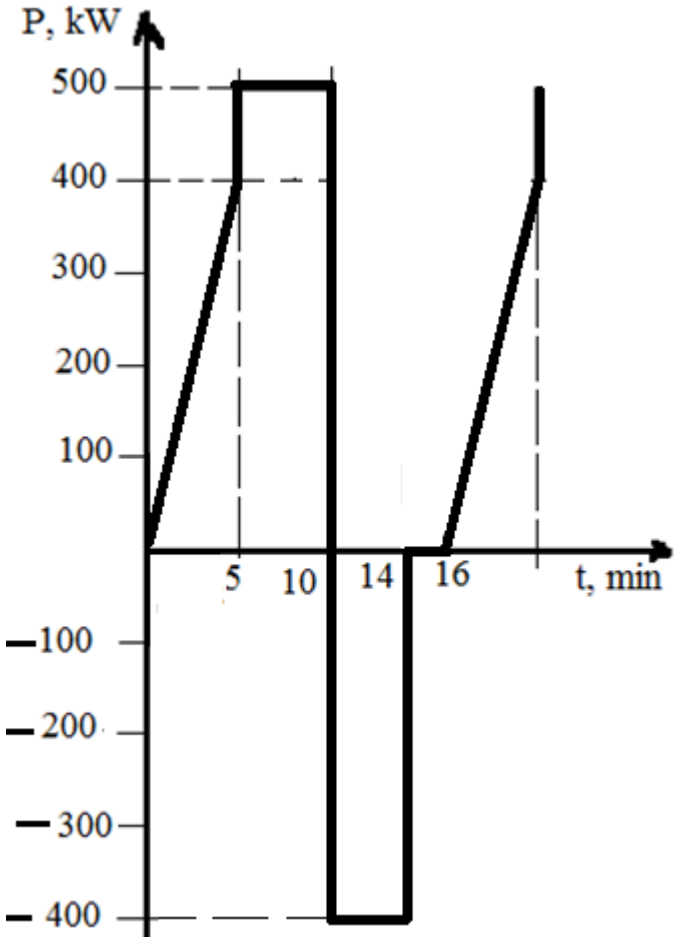
The motor is selected with $P_n \geq P_{eq}$ then checked for λ_T and starting torque T_{str}.

Notices:

- Through these three methods, the most accurate is the equivalent current method and it can be used for all cases except when the constant losses P_c are changing and the resistance R is changing such as in deep slot and double – cage squirrel cage motors where R is changing considerably at starting and braking.
- If the equivalent current method is not adequate, the equivalent torque method is not allowed. The equivalent torque is only used when the flux is constant in DC motors or cosϕ is constant in AC motors. Therefore, it can't be used in series DC motors and squirrel cage motors under stating and braking conditions.
- The equivalent power method can't be used if the motor speed is changing considerably under 'load and at starting and braking conditions.

Example:

A constant speed drive has the following duty cycle.
i) Load rising from 0 to 400 KW ; 5 min
ii) Uniform load of 500 KW; 5 min
iii) Regenerative power of 400 KW returned to the supply; 4 min
iv) Remains idle (rest) for ; 2 min
calculate the equivalent power.



Solution:

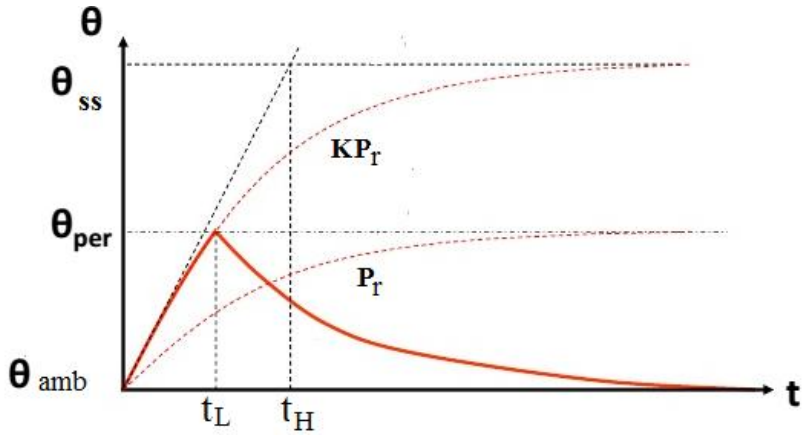
$$P_1 = \frac{400}{5}t, kW \qquad P_{1rms} = \sqrt{\frac{1}{t_1}\int_0^{t_1}(P_1)^2\,dt} = \sqrt{\frac{1}{5}\int_0^5\left(\frac{400}{5}t\right)^2 dt} = \frac{400}{\sqrt{3}}, kW$$

$$P_{eq} = \sqrt{\frac{\left(\frac{400}{\sqrt{3}}\right)^2 \times 5 + (500)^2 \times 5 + (400)^2 \times 4 + (0)^2 \times 2}{5+5+4+2}} = 367, kW$$

Short time duty

-In short time duty, time of motor operation is considerably less than the heating time constant and motor is allowed to cool down to the ambient temperature before it is required to operate again.

- A motor with continuous duty power rating P_r can be overloaded by an overload coefficient K (K>1) such that the power rating becomes KP_r and the maximum temperature rise reaches the permissible value (θ_{per}) for a short time duty T_L. So, ($P_{sh.t}=KP_r$).



When the duration of running period in a duty cycle with power KPr is t_L, then $\theta_{per} = \theta_{ss}(1 - e^{-t_L/T_H})$. So, $\frac{\theta_{ss}}{\theta_{per}} = \frac{1}{1-e^{-t_L/T_H}}$

Note that θ_{ss} is the steady state temperature rise which will be attained if the motor delivers a power (KP_r) on continuous bases, whereas the permissible temperature rise θ_{per} is also the steady state temperature rise attained when the motor operates with a Power P_r on continuous bases. The relation between motor losses during the short time duty and the rated losses is called heating overload coefficient(K_H). So,

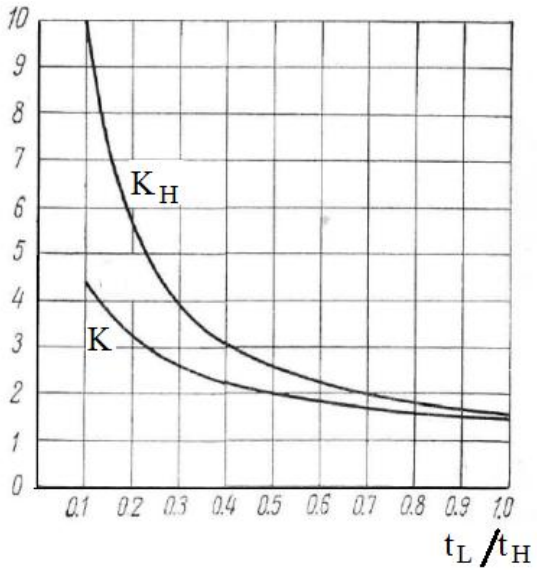
$$K_H = \frac{P_c + P_{cu}(\frac{P_{sh.t}}{P_r})^2}{P_c + P_{cu}} = \frac{\alpha + K^2}{\alpha + 1}$$

Where $\alpha = P_c/P_{cu}$ – ratio of constant to the variable losses (copper losses).
Since motor losses in short time duty is proportional to θ_{ss} and rated losses is proportional to θ_{per} , then

$$K_H = \frac{\theta_{ss}}{\theta_{per}} = \frac{1}{1 - e^{-t_L/T_H}} = \frac{\alpha + K^2}{\alpha + 1}. So, K = \sqrt{K_H(\alpha + 1) - \alpha} =$$

$$\sqrt{\frac{\alpha + 1}{1 - e^{-t_L/T_H}} - \alpha}$$

The last equation allows the calculation of overloading factor K which can be calculated when constant and variable losses are known separately. If not Known separately, total loss is assumed to be only proportional to $(Power)^2$; i.e. alpha (α) is assumed to be 0. So, $K = \sqrt{K_H}$.
As already mentioned, K is subjected to the constraints imposed (فرضت)by maximum allowable current in case of dc motors and breakdown torque limitations in case of IM and SM.



Ex1: A motor has a heating time constant of 60s and cooling time constant of 90s, if the motor rating is 20kw, find the permitted load power that can be used for 10s to let the motor work at its optimal utilization.

Solution:

If we neglect the constant losses, then the overload coefficient

$K = \sqrt{K_H} = \sqrt{\frac{1}{1-e^{-t_L/T_H}}} = \sqrt{\frac{1}{1-e^{-10/60}}} = 2.55$, since $P_{sh.t}=kP_r$, then $P_{sh.t}=2.55\times20=51$ kw.

Ex2: A motor has a heating time constant of 90s and a power of 60kw. The constant losses consist 80% of the copper losses when it works at rated power. The motor is working on short duty to drive a load for 30s. Find the permitted load power.

Solution:

The heating overload coefficient

$$K_H = \frac{1}{1-e^{-t_L/T_H}} = \frac{1}{1-e^{-30/90}} = 3.53$$

$$K_H = \frac{P_c + P_{cu}(\frac{P_{sh.t}}{P_r})^2}{P_c + P_{cu}} = \frac{0.8P_{cu} + P_{cu}(\frac{P_{sh.t}}{P_r})^2}{0.8P_{cu} + P_{cu}} = \frac{P_{cu}[0.8 + \left(\frac{P_{sh.t}}{P_r}\right)^2]}{1.8P_{cu}}$$

$$K_H = 3.53 = \frac{0.8 + \left(\frac{P_{sh.t}}{P_r}\right)^2}{1.8} = \frac{0.8 + \left(\frac{P_{sh.t}}{60}\right)^2}{1.8} \rightarrow P_{sh.t} = 141kw$$

INTERMITTENT PERIODIC DUTY

- During a Period of Operation, if the speed changes in wide limits, leading to Changes in heating and Cooling Conditions, methods of equivalent current, torque or power methods can't be employed.
- Consider a simple intermittent load, where the motor is alternately subjected to a fixed magnitude load P_r' of duration (t_r) and stand still condition of duration (t_s) as shown in diagram.

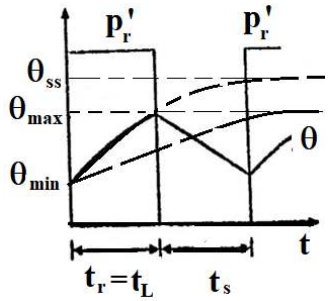


Figure ( ) Intermittent periodic load

- As motor is subjected to a periodic load, often the thermal steady- state is reached, the temperature rise will fluctuate between a maximum value (θ_{max}) and a minimum value (θ_{min}).
- For this load, the motor rating should selected such that ($\theta_{max} < \theta_{per}$) , where θ_{per} is maximum permissible temperature rise of the motor.

From temperature rise equation we can get

$\theta_{max} = \theta_{ss}(1 - e^{-\frac{t_r}{T_r}}) + \theta_{min}e^{-\frac{t_r}{T_r}}$ ……(1)

Where $\theta_{ss} \rightarrow$ temperature rise while ($P_r^{'}$) load applied continuously.

$\theta_{min} = (0)\left(1 - e^{-\frac{t_s}{T_s}}\right) + \theta_{max}e^{-\frac{t_s}{T_s}} = \theta_{max}e^{-\frac{t_s}{T_s}}$ ……(2)

with force cooling for continuous stand still, temperature rise goes to zero.

Substitute eq. (2) in eq.(1) yields

$\theta_{max} = \theta_{ss}\left(1 - e^{-\frac{t_r}{T_r}}\right) + \theta_{max}e^{-\frac{t_r}{T_r}}e^{-\frac{t_s}{T_s}}$ then,

$\theta_{ss}\left(1 - e^{-\frac{t_r}{T_r}}\right) = \theta_{max}\left[1 - e^{-\left(\frac{t_s}{T_s}+\frac{t_r}{T_r}\right)}\right]$ So,

$\frac{\theta_{ss}}{\theta_{max}} = \frac{1-e^{-\left(\frac{t_s}{T_s}+\frac{t_r}{T_r}\right)}}{1-e^{-\frac{t_r}{T_r}}}$……(3)

- $\frac{\theta_{ss}}{\theta_{max}}$ will be proportional to losses that would take place for two values of load and for full utilization ($\theta_{max} = \theta_{per}$) .
- If losses for load values (P_r) and ($P_r^{'}$) be denoted by (P_{1r}) and (P_{1s}) then,

$\frac{\theta_{ss}}{\theta_{max}} = \frac{P_{1s}}{P1r} = \frac{\propto+k^2}{\propto+1}$ ……(4)

where, $\alpha = \frac{const.losses}{copper\ losses} = \frac{P_c}{P_{cu}}$

From eq. (3) & (4), we get

$\propto +k^2 = (\propto +1)\left[\frac{1-e^{-\left(\frac{t_s}{T_s}+\frac{t_r}{T_r}\right)}}{1-e^{-\frac{t_r}{T_r}}}\right]$, then $k = \sqrt{(\propto +1)\left[\frac{1-e^{-\left(\frac{t_s}{T_s}+\frac{t_r}{T_r}\right)}}{1-e^{-\frac{t_r}{T_r}}}\right] - \propto}$…….(5)

where, $k = \frac{Pr'}{Pr}$ - overloading factor

t_r -running time, t_s – stand still time.

- If constant and copper losses are not available separately the α=0 and

$k = \sqrt{\frac{1-e^{-\left(\frac{t_s}{T_s}+\frac{t_r}{T_r}\right)}}{1-e^{-\frac{t_r}{T_r}}}}$……..(6)

- K is subjected to the constraints imposed by maximum allowable current in case of dc motors and breakdown torque limitations in case of IM and SM.

EXAMPLE 4.4

A motor has a heating time constant of 60 min and cooling time constant of 90 min. When run continuously on full load of 20 kW, the final temperature rise is 40℃.

1. What load motor can deliver for 10 min if this is followed by a shunt down period long enough for it to cool?
2. If it is on an intermittent load of 10 min followed by 10 min shut down, what is the maximum value of load it can supply during the on load period?

Sol: Given data,

Motor heating time Constant (τ) as (τ_r) = 60 min.

Motor cooling time Constant (τ_s) = 90 min.

Continuously full load (P_r) = 20k W.

Final temperature rise with continuous load (θ) =40°C.

(1) Motor running time (t) = 10min.

(2) With intermittent load.

Running time (tr) = 10min.

Shut down Stand still time (ts) = l0min.

On this case, constant losses are not Known Separately, hence α=0.

Maximum load in case (1) and in case (2) =?

In the first case, short time duty:

Overloading factor $k = \sqrt{\frac{1}{1-e^{-\frac{t}{\tau}}}} = \sqrt{\frac{1}{1-e^{-\frac{10}{60}}}} = 2.55$

The over loading factor $k = \frac{P_{max}}{P_r}$

So, $2.55 = \frac{P_{max}}{20\times10^3}$ then $P_{max} = 51 \times 10^3$watt

Case (2):

The over loading factor $k = \sqrt{\frac{1-e^{-[\frac{tr}{\tau r}+\frac{ts}{\tau s}]}}{1-e^{-[\frac{tr}{\tau r}]}}} = \sqrt{\frac{1-e^{-[\frac{10}{60}+\frac{10}{90}]}}{1-e^{-[\frac{10}{60}]}}} = \sqrt{\frac{0.2425}{0.1535}} = 1.2569$

Over loading factor $k = \frac{P_{max}}{P_r}$

$1.2569 = \frac{P_{max}}{20}$ S0, $P_{max} = 20 \times 1.257 = 25.14\ kw$

Resume: for the given drive:

Maximum load during short time duty = 51kw

Maximum load during intermittent duty = 25.14kw

EXAMPLE 4.5

Half hour (30min) rating of a motor is 100 kW. Heating time constant is 80 min and the maximum efficiency occurs at 70% full load. Determine the continuous rating of the motor.

Sol: Given data,

Motor running time (t_r) = 30min.

Motor heating time Constant (T_r) = 80 min.

Maximum efficiency occurs at load = 70% of full load=0.7 of full load

Continues rating of motor P=?

In this case, $t_r < \tau_r$ hence it is a short time duty. Short time duty max load $P_{max} = 100\ kw$.

Let Consider the full load Copper losses = P_{cu}.

Copper losses at 70% of full load $(P_{cu})_{0.7} = (0.7)^2 \times P_{cu} = 0.49P_{cu}$

At maximum efficiency:

Copper losses = constant losses.

$0.49 \times P_{cu} = P_c$ So, $P_{cu} = \frac{P_c}{0.49}$

The losses factor $\propto = \frac{P_c}{P_{cu}} = \frac{P_c}{\frac{P_c}{0.49}} = 0.49$

For short duty period, the over load factor $k = \sqrt{\frac{1+\alpha}{1-e^{-t_r/\tau}} - \alpha} = \sqrt{\frac{1+0.49}{1-e^{-\frac{30}{80}}} - 0.49} = 2.0672$

The over loading factor $k = \frac{P_{max}}{P_r}$, So,

$2.0672 = \frac{100}{P_r}$ then $P_r = \frac{100}{2.0672} = 48.37\text{kw}$

Resume: the continuous rating of motor (P_r) = 48.37kw